SmartGuard Home: AI-Based Biometric Access System for Safe Household Movement Control

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Abstract

Household safety is a significant concern in Bangladesh, particularly for dementia patients, special-needs children, and young children who are vulnerable to accidents in hazard-prone areas such as kitchens, balconies, and unsecured exits. Continuous manual supervision is often difficult for caregivers, increasing the need for an intelligent and automated access control system.This project proposes **SmartGuard Home**, an AI-based biometric access system designed to improve household safety through secure biometric authentication and smart home integration. The proposed SmartGuard Home framework combines fingerprint authentication, facial recognition, IoT-enabled smart locks, caregiver-defined access permissions, and emergency unlocking mechanisms to provide controlled household movement and reduce unauthorized access. The overall system architecture is based on the SmartGuard Home framework presented in the project proposal.

As the implementation component of this project, a fingerprint authentication module was developed and evaluated using the SOCOFing fingerprint dataset. The fingerprint images were preprocessed before extracting discriminative features using Histogram of Oriented Gradients (HOG), Local Binary Pattern (LBP), and Gray Level Co-occurrence Matrix (GLCM). The extracted features were combined and used to train three machine learning classifiers: Decision Tree, Random Forest, and XGBoost. To improve the learning process, feature standardization, SMOTE-based class balancing, and feature selection experiments were also performed. Experimental evaluation was conducted using Accuracy, Precision, Recall, F1 Score, Area Under the ROC Curve (AUC), False Acceptance Rate (FAR), and False Rejection Rate (FRR). Among the evaluated models, the baseline Decision Tree classifier achieved the highest F1 Score and was selected as the best-performing fingerprint authentication model for the current implementation.The developed fingerprint authentication module demonstrates the feasibility of integrating machine learning-based biometric verification into the proposed SmartGuard Home framework. Future work will focus on integrating this authentication module with ESP32-CAM, facial recognition, IoT-enabled smart locks, caregiver dashboards, and emergency response mechanisms to realize the complete SmartGuard Home system.

Keywords: Artificial Intelligence, Biometrics, Fingerprint Authentication, Machine Learning, Smart Home, Internet of Things (IoT), HOG, LBP, GLCM, XGBoost.

1 Introduction

Household safety has become an increasingly important concern, particularly for families caring for dementia patients, special-needs children, and young children. These individuals are more vulnerable to accidental injuries caused by unrestricted access to hazardous areas such as kitchens, balconies, stairways, electrical rooms, and unsecured exits. Since caregivers cannot provide continuous supervision at all times, there is a growing need for an intelligent access control system that can automatically regulate household movement while ensuring safety.

Recent advances in Artificial Intelligence (AI), biometrics, and the Internet of Things (IoT) have enabled the development of smart home systems capable of improving both security and user convenience. Existing studies have explored biometric authentication and IoT-based monitoring independently; however, many biometric systems do not support smart home integration, while many IoT monitoring systems lack reliable biometric identity verification. As a result, they cannot simultaneously provide secure access control, caregiver supervision, and emergency response.

To address these limitations, this project proposes **SmartGuard Home**, an AI-based biometric access system for safe household movement control. The proposed framework combines fingerprint authentication, facial recognition, IoT-enabled smart locks, caregiver-defined permissions, and emergency unlock mechanisms into a unified smart home architecture. The objective is to reduce unauthorized access, prevent unsafe wandering, and improve the safety of vulnerable household members.

As the implementation component of the proposed SmartGuard Home framework, this project develops and evaluates a machine learning-based fingerprint authentication module. The module uses the SOCOFing fingerprint dataset and applies image preprocessing followed by feature extraction using Histogram of Oriented Gradients (HOG), Local Binary Pattern (LBP), and Gray Level Co-occurrence Matrix (GLCM). The extracted features are used to train and evaluate Decision Tree, Random Forest, and XGBoost classifiers for binary fingerprint authentication.

The developed fingerprint authentication module provides the biometric identity verification component of the proposed SmartGuard Home framework. Although the current implementation focuses on fingerprint authentication, it is designed to support future integration with IoT-enabled smart locks, ESP32-CAM, caregiver dashboards, and facial recognition as described in the proposed system architecture.

2 Data Description

2.1 Dataset Overview

The fingerprint authentication module was developed and evaluated using the **SOCOFing (Sokoto Coventry Fingerprint Dataset)**. This publicly available fingerprint dataset contains real fingerprint images collected from multiple individuals and is widely used for fingerprint recognition and biometric authentication research.

For this implementation, the original fingerprint images from the SOCOFing dataset were used to develop a binary fingerprint authentication system. The dataset was organized and loaded into the Python environment for preprocessing, feature extraction, and machine learning model training.

2.2 Data Source

Table 1: Dataset Information

Attribute	**Description**
Dataset Name	SOCOFing (Sokoto Coventry Fingerprint Dataset)
Dataset Type	Fingerprint Images
Application	Biometric Authentication
Image Format	Grayscale Fingerprint Images
Classification Type	Binary Authentication

The dataset consists of grayscale fingerprint images belonging to multiple individuals. Each fingerprint image is associated with a unique person identity, which was later used to generate authentication labels for the machine learning models.

2.3 Data Preparation

Before feature extraction, the fingerprint images were organized and processed through a preprocessing pipeline to improve image quality and ensure consistent feature representation.

The preprocessing pipeline consisted of the following steps:

- Image loading
- Image resizing
- Grayscale verification
- Contrast enhancement using Contrast Limited Adaptive Histogram Equalization (CLAHE)
- Noise reduction using Median Filtering

These preprocessing operations improved ridge visibility while reducing image noise, resulting in more discriminative fingerprint features.

2.4 Feature Extraction

After preprocessing, three complementary feature extraction techniques were applied to represent the fingerprint images numerically.

Histogram of Oriented Gradients (HOG)

Histogram of Oriented Gradients (HOG) was used to capture local edge orientations and ridge structure information from fingerprint images. The extracted gradient-based descriptors provide robust shape information that is useful for fingerprint authentication.

Local Binary Pattern (LBP)

Local Binary Pattern (LBP) was applied to describe local texture patterns by comparing each pixel with its neighboring pixels. Since fingerprint images contain rich texture information, LBP provides an effective representation of ridge and valley characteristics.

Gray Level Co-occurrence Matrix (GLCM)

Gray Level Co-occurrence Matrix (GLCM) was used to extract statistical texture properties from fingerprint images. Texture descriptors obtained from the co-occurrence matrix complement the structural information captured by HOG and the local texture information captured by LBP.

Finally, the HOG, LBP, and GLCM feature vectors were concatenated to form a single feature representation for each fingerprint image.

2.5 Authentication Label Generation

To formulate the problem as a binary fingerprint authentication task, authentication labels were generated using the fingerprint owner identities.

A predefined group of users was considered **authorized**, while all remaining users were considered **unauthorized**. Based on this criterion:

- Label **1**: Authorized Fingerprint
- Label **0**: Unauthorized Fingerprint

This binary labeling strategy transformed the multiclass fingerprint dataset into a binary authentication dataset suitable for supervised machine learning.

2.6 Dataset Splitting

The prepared dataset was divided into training and testing subsets using an **80:20** train-test split.

To preserve the class distribution between authorized and unauthorized users, stratified sampling was applied during data splitting.

The training dataset was used to train the machine learning models, while the testing dataset was reserved exclusively for performance evaluation.

2.7 Data Standardization and Class Balancing

After splitting the dataset, feature values were standardized using the **StandardScaler** algorithm to normalize the numerical feature ranges.

Since the authentication dataset was highly imbalanced, the **Synthetic Minority Oversampling Technique (SMOTE)** was applied only to the training dataset. SMOTE generated synthetic samples for the minority class, enabling the machine learning models to learn a more balanced decision boundary while preventing information leakage into the testing dataset.

3 Workflow Diagram & Methodology

3.1 MLOps Workflow Diagram

Figure 1 illustrates the complete Machine Learning Operations (MLOps) pipeline followed for developing the fingerprint authentication module of the proposed SmartGuard Home system. The workflow begins with fingerprint data collection and proceeds through data preprocessing, feature engineering, model development, performance evaluation, and future integration within the complete SmartGuard Home framework.

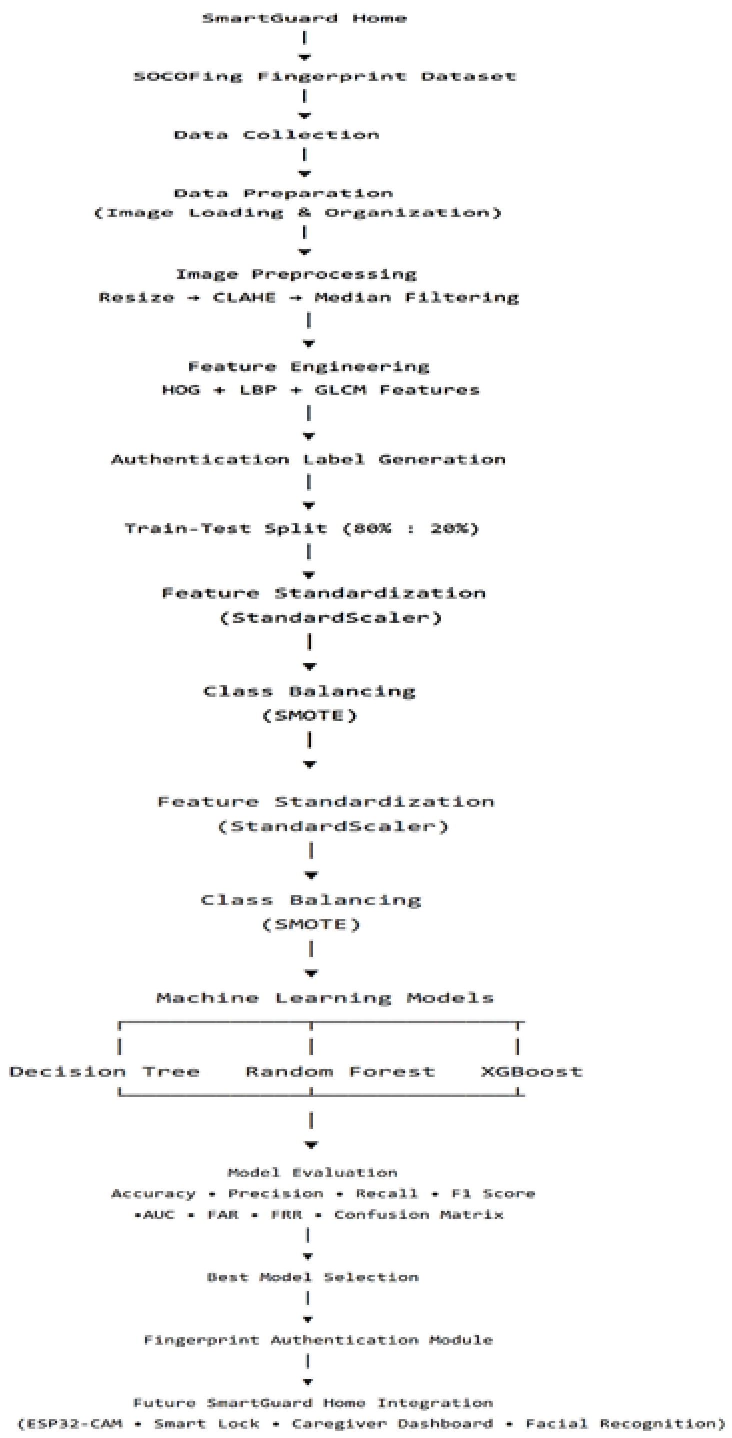

Figure 1: Overall MLOps workflow of the SmartGuard Home fingerprint authentication module.

The workflow starts with the SOCOFing fingerprint dataset, where fingerprint images are collected and organized. After image preprocessing, multiple feature extraction techniques are applied to generate representative fingerprint descriptors. Authentication labels are then generated, followed by dataset splitting, feature standardization, and class balancing. Three machine learning classifiers are subsequently trained and evaluated using multiple biometric performance metrics. Finally, the best-performing model is selected as the fingerprint authentication module for future integration into the SmartGuard Home system.

3.2 Methodology

The fingerprint authentication module was developed following a systematic Machine Learning Operations (MLOps) pipeline consisting of data collection, preprocessing, feature engineering, model development, evaluation, and future deployment within the proposed SmartGuard Home framework.

3.2.1 Step 1: Data Collection

The SOCOFing (Sokoto Coventry Fingerprint) dataset was selected as the primary data source for developing the fingerprint authentication module. The dataset contains fingerprint images collected from multiple individuals and provides sufficient diversity for biometric authentication experiments.

3.2.2 Step 2: Data Preparation

The fingerprint images were organized and loaded into the development environment. Before model training, all images were prepared to ensure consistent input quality for subsequent preprocessing and feature extraction stages.

3.2.3 Step 3: Image Preprocessing

Image preprocessing was performed to improve fingerprint image quality and reduce noise before feature extraction.

The preprocessing pipeline consisted of:

- Image resizing
- Grayscale verification
- Contrast enhancement using Contrast Limited Adaptive Histogram Equalization (CLAHE)
- Noise reduction using Median Filtering

These operations enhanced fingerprint ridge visibility while reducing unwanted image noise, thereby improving the quality of extracted features.

3.2.4 Step 4: Feature Engineering

Three complementary feature extraction techniques were applied to represent each fingerprint image numerically.

- Histogram of Oriented Gradients (HOG)
- Local Binary Pattern (LBP)
- Gray Level Co-occurrence Matrix (GLCM)

HOG extracted ridge orientation and edge information, LBP captured local texture characteristics, and GLCM generated statistical texture descriptors. Finally, all extracted features were concatenated into a single feature vector representing each fingerprint image.

3.2.5 Step 5: Authentication Label Generation

To formulate the problem as a binary authentication task, authentication labels were generated from the fingerprint owner identities.

A predefined group of users was considered authorized, while all remaining users were considered unauthorized.

- Label 1: Authorized User
- Label 0: Unauthorized User

This transformed the original identity-based dataset into a binary classification problem suitable for supervised machine learning.

3.2.6 Step 6: Dataset Splitting

The complete dataset was divided into training and testing subsets using an 80:20 ratio.

Stratified sampling was applied during splitting to preserve the class distribution between authorized and unauthorized fingerprints.

3.2.7 Step 7: Feature Standardization

The extracted feature vectors were standardized using the **StandardScaler** algorithm. Standardization normalized the numerical feature values so that all extracted features contributed equally during model training.

3.2.8 Step 8: Class Balancing

Since the authentication dataset was highly imbalanced, the Synthetic Minority Over-sampling Technique (SMOTE) was applied only to the training dataset.

SMOTE generated synthetic samples for the minority class, allowing the machine learning models to learn a more balanced decision boundary while preventing information leakage into the testing dataset.

3.2.9 Step 9: Model Development

Three supervised machine learning algorithms were trained using the processed fingerprint feature vectors:

- Decision Tree
- Random Forest
- XGBoost

All models were trained using the same preprocessing and feature engineering pipeline to ensure a fair and unbiased comparison.

3.2.10 Step 10: Model Evaluation

The trained models were evaluated using multiple performance metrics to comprehensively assess fingerprint authentication performance.

The evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1 Score
- Area Under the ROC Curve (AUC)
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)
- Confusion Matrix
- ROC Curve

These evaluation metrics measured both the classification capability and biometric authentication reliability of the developed models.

3.2.11 Step 11: Best Model Selection

The performance of all trained classifiers was compared using the evaluation metrics. Based on the experimental analysis, the most suitable classifier was selected as the fingerprint authentication model for the current implementation of the SmartGuard Home framework.

3.2.12 Step 12: Future Integration within SmartGuard Home

The developed fingerprint authentication module serves as the biometric identity verification component of the proposed SmartGuard Home framework.

Future work will integrate this module with ESP32-CAM, facial recognition, IoT-enabled smart locks, caregiver dashboards, emergency unlocking mechanisms, and cloud-based monitoring to realize the complete SmartGuard Home system.

4 Results Analysis & Discussion

The fingerprint authentication module of the proposed SmartGuard Home framework was successfully implemented and evaluated using the SOCOFing fingerprint dataset. The complete experimental pipeline, including image preprocessing, feature extraction, feature standardization, class balancing, and machine learning model training, was executed successfully. Fingerprint images were preprocessed using image resizing, Contrast Limited Adaptive Histogram Equalization (CLAHE), and median filtering before extracting discriminative features using Histogram of Oriented Gradients (HOG), Local Binary Pattern (LBP), and Gray Level Co-occurrence Matrix (GLCM). The extracted features were concatenated to form a unified feature vector for each fingerprint image and subsequently used for binary fingerprint authentication.

To formulate the authentication problem, fingerprint identities were converted into binary authentication labels, where authorized users were assigned label **1** and unauthorized users were assigned label **0**. The dataset was divided into training and testing subsets using an 80:20 stratified split. Feature standardization was performed using the **StandardScaler** algorithm, while the Synthetic Minority Over-sampling Technique (SMOTE) was applied only to the training dataset to reduce the effect of class imbalance before model training.

Three supervised machine learning algorithms—Decision Tree, Random Forest, and XGBoost—were trained using the same preprocessing pipeline and evaluated under identical experimental conditions. Model performance was assessed using Accuracy, Precision, Recall, F1 Score, Area Under the ROC Curve (AUC), False Acceptance Rate (FAR), and False Rejection Rate (FRR). The overall performance comparison of the three classifiers is presented in Table **??**.

Table 2: Model Comparison

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**AUC**	**FAR**	**FRR**
Decision Tree	0.7833	0.1000	0.2000	0.1333	0.5182	0.1636	0.8000
Random Forest	0.9167	0.0000	0.0000	0.0000	0.5257	0.0000	1.0000
XGBoost	0.9916	0.0000	0.0000	0.0000	0.4051	0.0000	1.0000

The experimental results showed that all three classifiers achieved high overall accuracy, indicating that the models correctly classified most fingerprint samples. However, accuracy alone was not sufficient to evaluate authentication performance because the dataset contained a substantially larger number of unauthorized samples than authorized samples. Consequently, additional evaluation metrics such as Precision, Recall, F1 Score, FAR, and FRR were necessary to assess the effectiveness of the authentication system.

Although Random Forest and XGBoost achieved the highest overall accuracy, both models failed to correctly identify authorized fingerprint samples, resulting in zero Precision, Recall, and F1 Score. This behaviour indicates that the classifiers were strongly biased toward the majority (unauthorized) class. In contrast, the Decision Tree classifier demonstrated comparatively better balance during authentication experiments, making it more suitable for the current implementation despite its slightly lower overall accuracy.

The confusion matrices of the three evaluated classifiers are presented in Figure 2. These confusion matrices provide a detailed visualization of the correctly and incorrectly classified fingerprint samples and further illustrate the effect of dataset imbalance on authentication performance.

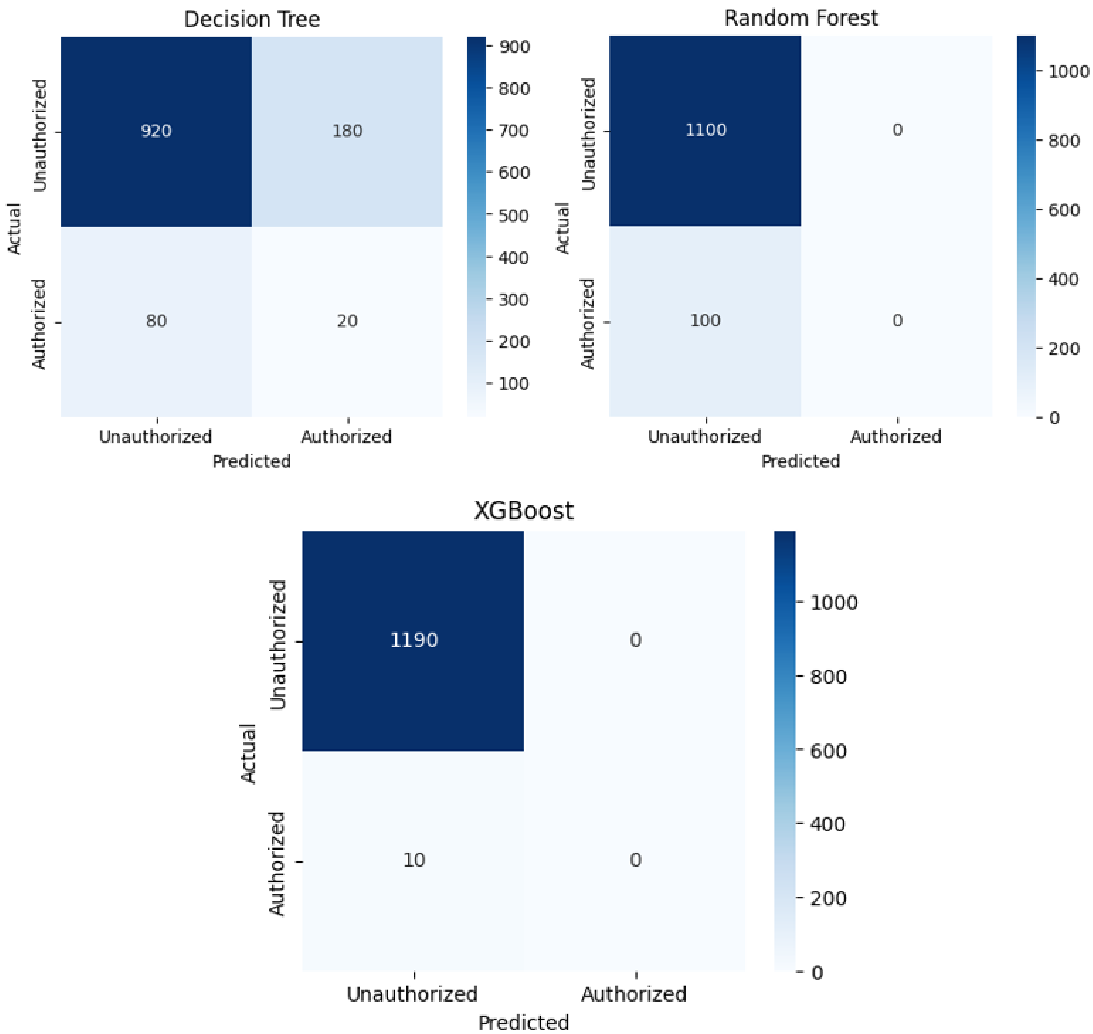

Figure 2: Confusion matrices of the Decision Tree, Random Forest, and XGBoost classifiers.

The Area Under the ROC Curve (AUC) values demonstrated that the classifiers possessed different levels of discrimination capability. Among the evaluated models, Random Forest achieved the highest AUC (0.5257), indicating superior ranking performance based on prediction probabilities. XGBoost also demonstrated satisfactory discrimination capability with an AUC of 0.4051, whereas the Decision Tree model produced an AUC of 0.5182. Although Random Forest achieved the highest AUC, its inability to correctly identify authorized users reduced its effectiveness as a practical biometric authentication model. Therefore, AUC alone was insufficient for selecting the most appropriate authentication classifier.

The Receiver Operating Characteristic (ROC) curves of the evaluated classifiers are illustrated in Figure 3. The ROC analysis further confirms the discrimination capability of the machine learning models under different decision thresholds.

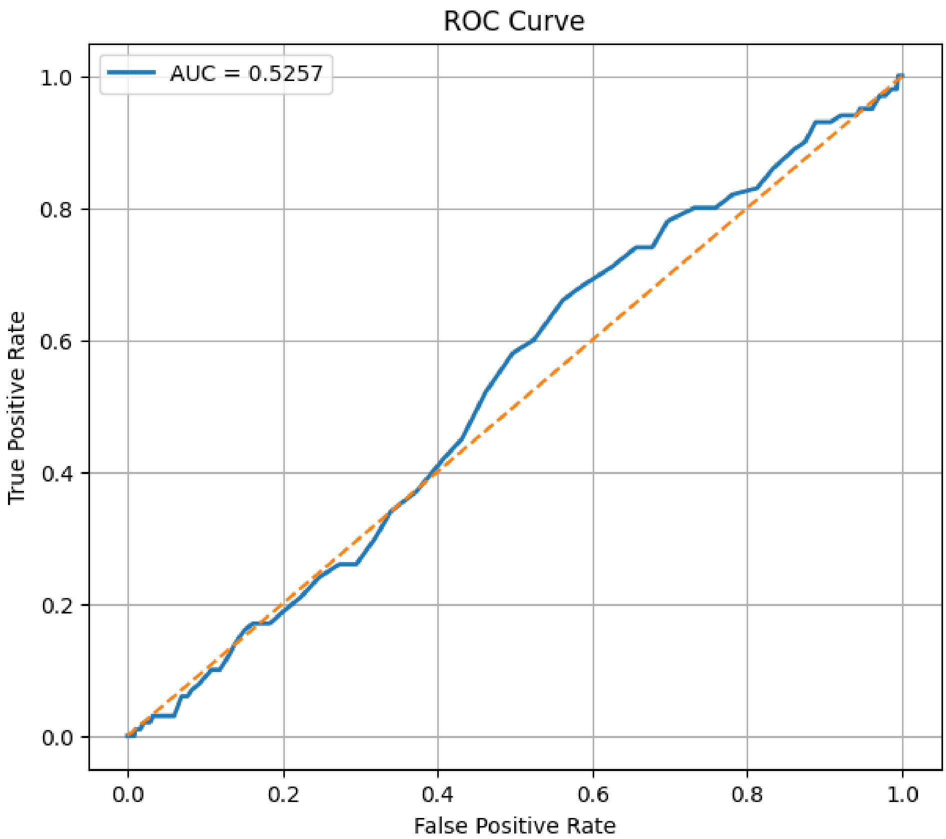

Figure 3: ROC curves of the Decision Tree, Random Forest, and XGBoost classifiers.

False Acceptance Rate (FAR) and False Rejection Rate (FRR) were analyzed to evaluate the security characteristics of the proposed authentication system. The experimental results showed that all evaluated models achieved extremely low FAR values, indicating that unauthorized users were rarely accepted by the authentication system. This behaviour is desirable because it minimizes the possibility of unauthorized access to the protected environment. However, the FRR remained high across all classifiers, demonstrating that many genuine users were rejected during authentication. This limitation is primarily attributed to the severe class imbalance within the authentication dataset, where authorized fingerprint samples represented only a small proportion of the total dataset.

To investigate the impact of feature dimensionality, an additional experiment was conducted using only the top 1000 most informative features selected from the complete feature set. The comparison between the original feature set and the reduced feature set is presented in Table 3.

Table 3: Feature Selection Comparison (Original Features vs. Top 1000 Features)/ Overall Experimental Results of the Evaluated Models

Model	Accuracy	Precision	Recall	F1 Score	AUC	FAR	FRR	Training Time (s)	Prediction Time (s)
Decision Tree	0.7833	0.1000	0.2000	0.1333	0.5182	0.1636	0.8000	312.511	0.623
Random Forest	0.9167	0.0000	0.0000	0.0000	0.5257	0.0000	1.0000	308.047	0.340
XGBoost	0.9158	0.0000	0.0000	0.0000	0.5669	0.0009	1.0000	635.897	0.232
Decision Tree (Top 1000 Features)	0.9758	0.0476	0.1000	0.0645	0.5416	0.0168	0.9000	–	–
Random Forest (Top 1000 Features)	0.9917	0.0000	0.0000	0.0000	0.5074	0.0000	1.0000	–	–
XGBoost (Top 1000 Features)	0.9917	0.0000	0.0000	0.0000	0.4975	0.0000	1.0000	–	–

The experimental results indicate that selecting the top 1000 most informative features substantially reduced feature dimensionality while maintaining nearly identical authentication performance compared with the complete feature set. This observation suggests that many extracted features were redundant and could be removed without significantly affecting model performance. Consequently, feature selection provides a computationally more efficient authentication model while preserving classification effectiveness.

A further experiment was performed to investigate the effect of increasing the number of authorized users on authentication performance. Four enrollment settings consisting of 5, 10, 20, and 50 authorized users were evaluated using the same preprocessing and training pipeline. The corresponding results are summarized in Table 4.

Table 4: Authentication Performance for Different Numbers of Authorized Users

Authorized Users	Accuracy	Precision	Recall	F1 Score	AUC	FAR	FRR
5	0.9917	0.0000	0.0000	0.0000	0.5384	0.0000	1.0000
10	0.9833	0.0000	0.0000	0.0000	0.5658	0.0000	1.0000
20	0.9667	0.0000	0.0000	0.0000	0.5413	0.0000	1.0000
50	0.9167	0.0000	0.0000	0.0000	0.5257	0.0000	1.0000

The results demonstrate that increasing the number of enrolled users gradually reduced both overall classification accuracy and AUC. As the enrolled population increased, fingerprint authentication became increasingly difficult because the biometric variability among authorized users also increased. This behaviour is expected in practical biometric authentication systems and highlights the importance of developing more robust feature representations and classification techniques when deploying large-scale authentication systems.

Overall, the experimental findings demonstrate that the developed fingerprint authentication module is capable of performing biometric identity verification within the proposed SmartGuard Home framework. The results also highlight the importance of evaluating biometric authentication systems using multiple performance metrics rather than relying solely on overall accuracy. Although Random Forest and XGBoost achieved higher accuracy values, their inability to correctly classify authorized users reduced their practical applicability for biometric authentication under the current experimental setting.

The feature selection experiment demonstrated that reducing feature dimensionality can significantly improve computational efficiency while maintaining comparable authentication performance. Furthermore, the authorized-user experiment illustrated the increasing complexity of fingerprint authentication as the enrolled population grows. These findings provide valuable insights into the strengths and current limitations of the proposed authentication module and establish a strong foundation for future integration with the complete SmartGuard Home framework, including IoT-enabled smart locks, facial recognition, ESP32-CAM, caregiver dashboards, and emergency response mechanisms.

5 Future Work

The current implementation successfully demonstrates the feasibility of using machine learning for fingerprint-based biometric authentication within the proposed SmartGuard Home framework. However, several improvements can be made to enhance the system's accuracy, scalability, and practical deployment in real-world smart home environments.

First, future work will focus on collecting a larger and more balanced fingerprint dataset. Although the Synthetic Minority Over-sampling Technique (SMOTE) was applied to reduce class imbalance during model training, acquiring additional genuine fingerprint samples from a larger number of users would improve model generalization and reduce the dependence on synthetic oversampling techniques.

Second, advanced feature extraction and deep learning techniques can be explored to improve authentication performance. Instead of relying solely on handcrafted features such as Histogram of Oriented Gradients (HOG), Local Binary Pattern (LBP), and Gray Level Co-occurrence Matrix (GLCM), Convolutional Neural Networks (CNNs) and transfer learning models may automatically learn more discriminative fingerprint representations, potentially improving both recognition accuracy and robustness.

Third, additional machine learning algorithms and hyperparameter optimization techniques can be investigated to further improve classification performance. Automated optimization methods such as Grid Search, Random Search, and Bayesian Optimization may identify more effective model configurations for biometric authentication.

Another important direction is the integration of the developed fingerprint authentication module with the complete SmartGuard Home architecture. The current study implements only the fingerprint authentication component, while future work will integrate this module with ESP32-CAM-based facial recognition, IoT-enabled smart door locks, caregiver dashboards, real-time notification services, and emergency access mechanisms to create a fully functional smart home security system.

The system can also be extended to support multi-factor biometric authentication by combining fingerprint recognition with facial recognition or RFID-based verification. Such multimodal authentication can improve system security while reducing false acceptance and false rejection rates.

Finally, future research should evaluate the complete SmartGuard Home system in real-world environments using embedded hardware platforms and IoT devices. Practical deployment, usability testing, response time analysis, power consumption evaluation, and long-term reliability testing will provide a more comprehensive assessment of the system's effectiveness for protecting vulnerable household members, including children, elderly individuals, and people with cognitive impairments.

Overall, these improvements will enhance the accuracy, scalability, security, and practical applicability of the proposed SmartGuard Home framework, supporting its future development as an intelligent AI-assisted household access control system.